

Real Madrid CF

Tactical & Player Performance Dashboard

KPI, Metrics & Formula Reference

Definitions, calculation methods and key analytical considerations for every metric computed by the dashboard — sourced strictly from Opta Stats Perform event data.

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1 · Data Foundation & Pipeline

Every figure in the dashboard is derived from raw **Opta Stats Perform** event-level JSON match files. No synthetic, simulated or manually entered values are used. The pipeline parses each match's event stream into a normalised table, then aggregates upward to match-, season- and player-level KPIs.

Dataset scope: 108 clean match files · 3 competitions (LaLiga · UEFA Champions League · Copa del Rey) · 2 seasons (2024-25, 2025-26) · Real Madrid matches only. Files are cached with an LRU cache for performance.

Opta event type IDs used

typeld	Event	Used for
1, 2	Pass / Offside Pass	Possession, pass accuracy, pass networks
3	Take On / Dribble	Player dribble counts
4	Foul	Fouls committed
6	Corner Awarded	Corner counts
7	Tackle	Defensive actions, PPDA
8	Interception	Defensive actions, PPDA
13	Shot On Target (saved)	Shots, shots on target, xG
15	Shot Blocked	Shots, xG
16	Goal	Goals, shots on target, xG
19	Player On (sub)	Lineup status (Starter vs Sub-on)
49	Ball Recovery	Defensive actions, PPDA
65 / 68	Yellow / Red Card	Discipline counts

Qualifier IDs: 140/141 = pass end coordinates · 56 = pitch zone · 279 = pass direction · 396 = xGOT proxy (0-100) · 5/72 = set-piece restarts. Bookkeeping events (27, 28, 30, 32, 34 — period starts, formations) are filtered out.

2 · Match-Level KPIs

Computed per match by `calc_match_kpis()` over the normalised event table (RM = Real Madrid, OPP = opponent).

Metric	Formula / definition
Goals scored / conceded	Final score from matchInfo (RM goals, opponent goals).
Result (W/D/L)	Sign of (RM score – opponent score).
Possession %	$\text{RM passes} \div (\text{RM passes} + \text{OPP passes}) \times 100$. Pass-volume share proxy.
Pass accuracy %	$\text{mean}(\text{outcome}) \times 100$ over RM passes, where outcome = 1 if successful.
Shots (total)	Count of RM shot events: typeld $\in \{13, 15, 16\}$ or shot qualifier present.
Shots on target	Count of RM events with typeld $\in \{13, 16\}$ (saved attempt + goal).
xG for / against	$\sum$ positional-model xG over RM / OPP shot events (see §4).
Tackles / Interceptions / Recoveries	Count of RM events typeld 7 / 8 / 49.

Metric	Formula / definition
Corners / Fouls	Count of RM events typeld 6 / 4.
Yellow / Red cards	Count of RM events typeld 65 / 68.
PPDA (match KPI)	OPP passes with $x \geq 40 \div$ RM defensive actions (typeld 7,8,49).

3 · Season-Level Aggregated KPIs

Computed by `calc_season_kpis()`, which scans every RM match in a competition-season and aggregates the per-match KPIs above.

Metric	Formula / definition
Played, Wins, Draws, Losses	Counts of matches by result.
Win %	$\text{wins} \div \text{played} \times 100$.
Goals scored / conceded	Σ of per-match goals for / against.
Goal difference	Goals scored – goals conceded.
Goals per game	Goals scored $\div$ played.
Conceded per game	Goals conceded $\div$ played.
Avg possession %	Mean of per-match possession% (simple average across matches).
Avg pass accuracy %	Mean of per-match pass accuracy%.
xG for / against	Σ of per-match xG for / against.
xG difference	Season xG for – season xG against.

4 · Expected Goals (xG) — Positional Model

The dashboard uses a transparent, **calibrated positional xG model** based on shot location, not the provider's proprietary xG. It is applied to every shot event so xG is always present (the Opta post-shot qualifier 395 is deliberately excluded — it appears on only ~11% of shots, goals only, with unreliable bimodal values).

Penalties

Detected by Opta's standardised penalty-spot coordinate ($x = 88.5$, $y = 50.0$, ± 0.2) and assigned a fixed **xG = 0.76** (empirical penalty conversion rate).

Open-play logistic model

Pitch is Opta 0-100 units ($x \approx 105$ m, $y \approx 68$ m); attacking goal at $x = 100$, $y = 50$; goal half-width 3.66 m.

```
dx_m = (100 - x) × 1.05    dy_m = (y - 50) × 0.68
dist_m = √(dx_m² + dy_m²)
angle_rad = | atan2(dy_m - 3.66, dx_m) - atan2(dy_m + 3.66, dx_m) |
logit(xG) = -3.785 - 0.0337 · dist_m + 3.64 · angle_rad
xG = 1 / (1 + e-logit),    clipped to [0.01, 0.99]
```

Calibration anchors (central, open play): 6 m → ≈ 0.50 · 11 m → ≈ 0.14 · edge of box 16.5 m → ≈ 0.06 · long shot 25 m → ≈ 0.027 .

5 · Tactical Metrics

PPDA — Passes Allowed Per Defensive Action

Pressing-intensity metric (lower = more intense press). The tactical page uses `calc_ppda()`, restricting both terms to the pressing zone ($x \geq 40$):

```
PPDA = ( OPP passes with x ≥ 40 ) ÷ ( RM defensive actions [tackle, interception, recovery]
with x ≥ 40 )
```

Field Tilt

Share of final-third activity belonging to Real Madrid — a territorial dominance proxy.

```
Field tilt % = RM touches with x ≥ 67 ÷ all touches with x ≥ 67 × 100
```

6 · Tactical Phase Model (A / B / C / D)

Four play-phase scores, each normalised to 0-100, computed by `phase_scores_from_events()`. They blend weighted components so the radar/benchmark views compare phases on one scale.

Phase	Weighted formula (clipped 0-100)
A · Offensive Moment	$\text{pass_acc} \times 0.45 + (\Sigma xg \times 10) \times 0.35 + (\text{shots} \times 2.5) \times 0.20$
B · Defensive Transition	$(100 / (1 + \text{PPDA})) \times 0.55 + ((\text{recoveries} + \text{interceptions}) \times 1.8) \times 0.45$
C · Defensive Moment	$(100 / (1 + \text{PPDA})) \times 0.40 + \max(0, 40 - \text{opp_shots} \times 2) \times 0.60$
D · Offensive Transition	$\text{trans_rate} \times 0.65 + (\Sigma xg \times 7) \times 0.35$

trans_rate = % of ball regains (recovery / interception / tackle) that lead to a Real Madrid shot within 15 seconds. A parallel `phase_scores_from_team_row()` variant recomputes the same four phases from per-game season aggregates (goals_pg, xg_pg, shots_pg, def_actions_pg) for cross-competition benchmarking.

7 · Player-Level Metrics

Per-player aggregates from `get_player_stats()`, grouped by player across the selected scope; feed the squad table and multi-radar.

Metric	Definition
Minutes played	Max event minute for the player (proxy for time on pitch).
Passes / Pass accuracy	Pass count; $\text{mean}(\text{outcome}) \times 100$ over the player's passes.
Shots / Shots on target / Goals	Counts of the player's shot, SoT and goal events.
Key passes / Assists	Opta keyPass flag count; assist flag count.
Tackles / Interceptions / Recoveries / Dribbles	Counts of typeld 7 / 8 / 49 / 3.
xG	Σ positional-model xG over the player's shots.
Fouls	Count of the player's foul events (typeld 4).
Lineup status	typeld 19 $\Rightarrow$ 'Sub On'; otherwise 'Starting 11'.

8 · Key Points & Considerations

- **100% real data.** Every metric is derived from Opta event files; no synthetic or estimated values are introduced anywhere in the pipeline.
- **Possession is a pass-share proxy.** The event feed has no ball-possession duration, so possession% uses each side's share of total passes — directionally accurate, not a stopwatch figure.
- **xG is a transparent positional model** (distance + shot angle logistic), not the provider's proprietary xG. Penalties are flagged by standardised coordinate and fixed at 0.76. Opta qualifier 395 is excluded because it only covers goals (~11% of shots) with bimodal values.
- **PPDA pressing zone = $x \geq 40$.** Lower PPDA means a more aggressive press; the tactical page restricts both passes and defensive actions to this zone for consistency.
- **Phase scores are weighted composites** normalised to 0–100 for comparability — they rank relative phase strength, they are not physical units.
- **Lineup status is inferred** from substitution-on events (typeld 19); all other players with events are treated as starters.
- **Season averages are match-simple means** (e.g. avg possession), not minutes- or events-weighted, keeping each match an equal observation.
- **Data hygiene.** Only Real-Madrid matches are loaded, bookkeeping events are stripped, files are LRU-cached, and the canonical clean bundle (dashboard/data, 0 corrupt files) is the single source of truth.
- **Optional advanced layer.** An experimental tactical-intelligence overlay (sub-phase tagging, advanced/cross-phase KPIs, sequence enrichment) exists behind feature flags that default to OFF, so it never affects the standard metrics above unless explicitly enabled via environment variables.